Aug. 3, 1926. 1,594,345

H. E. BAKKEN ET AL

PRODUCTION OF MAGNESIUM

Filed May 8, 1923 3 Sheets-Sheet 1

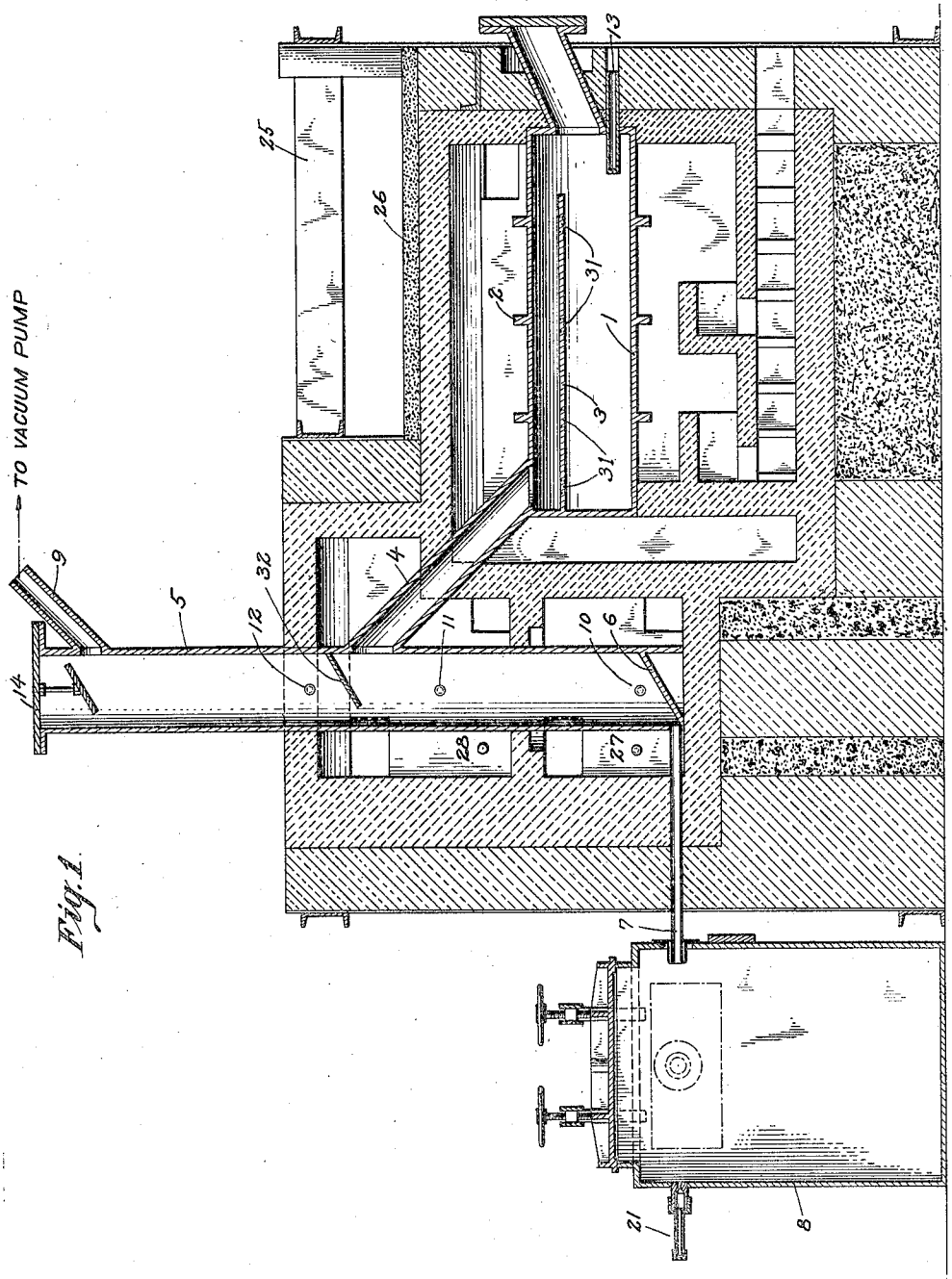


Fig. 1.

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Filed May 8, 1923 3 Sheets-Sheet 2

Fig. 4.

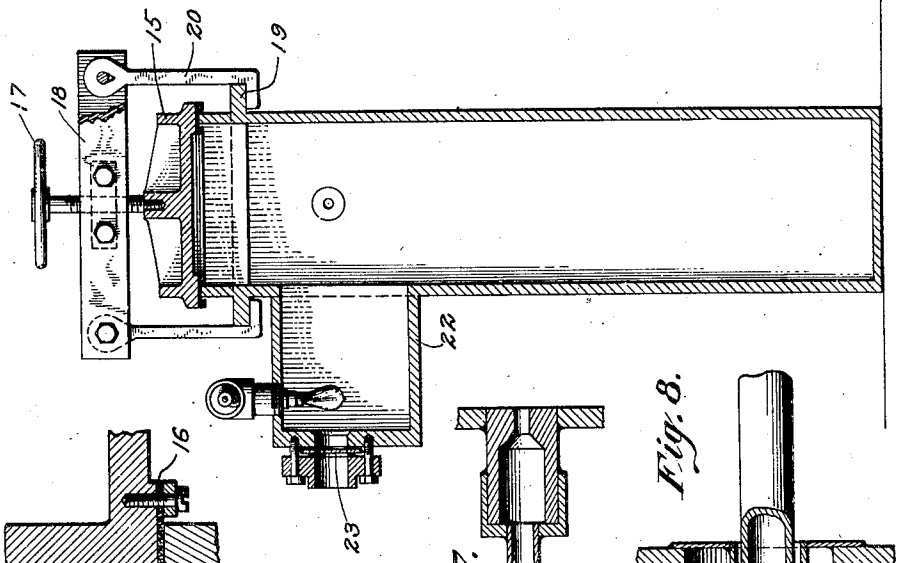


Fig. 6.

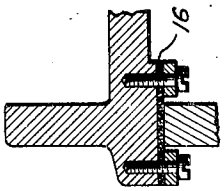


Fig. 7.

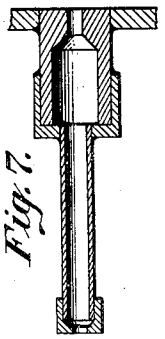


Fig. 8.

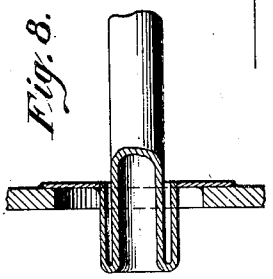


Fig. 2.

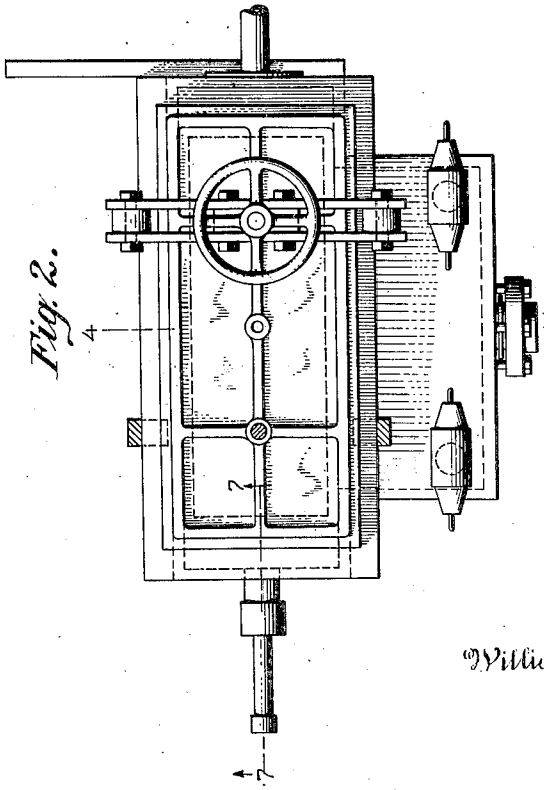
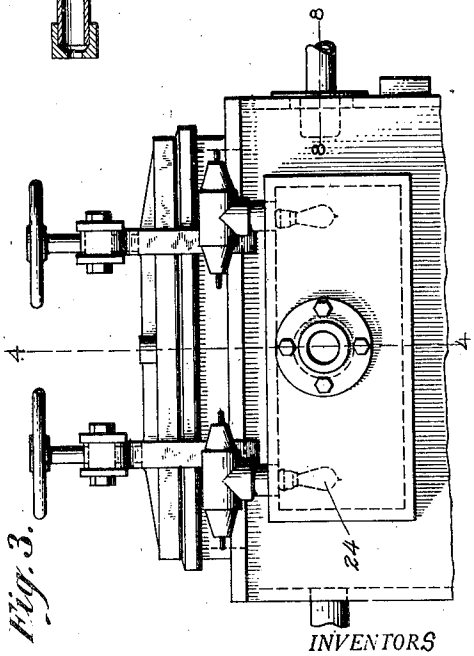


Fig. 3.



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Filed May 8, 1923 3 Sheets-Sheet 3

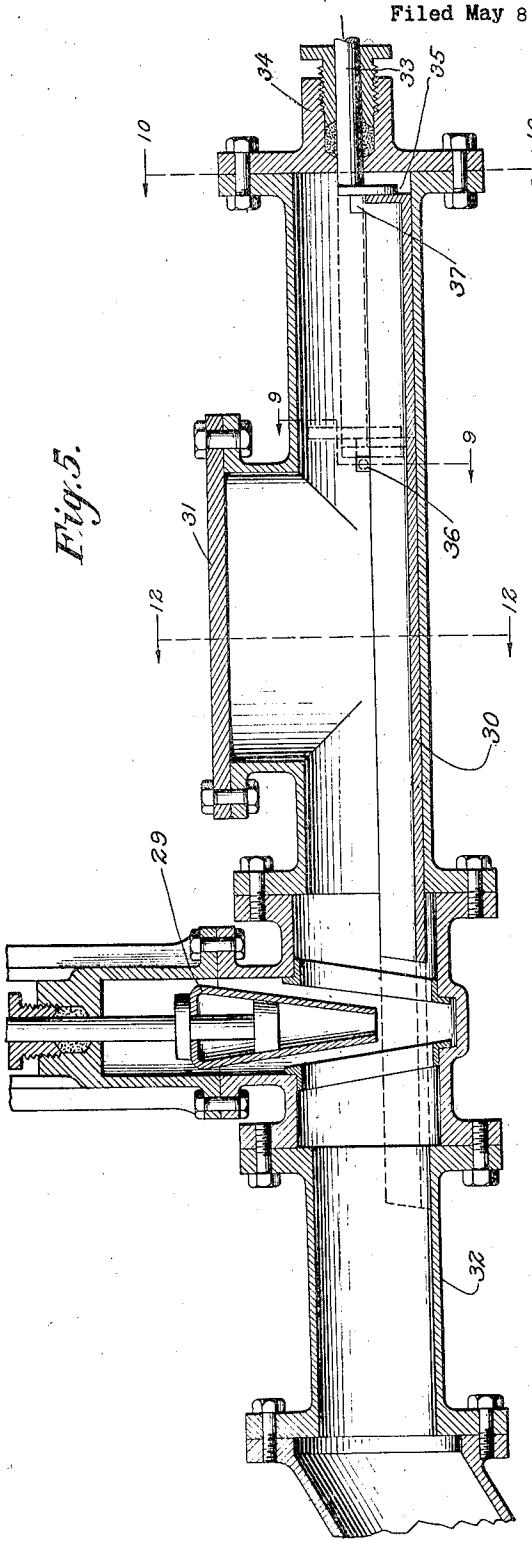


Fig. 5.

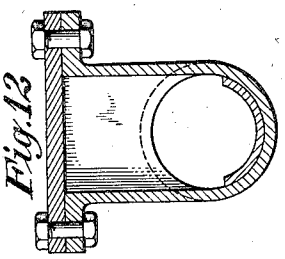


Fig. 12.

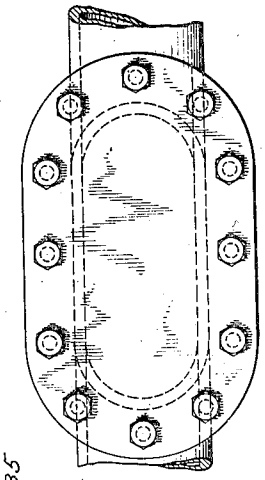


Fig. 11.

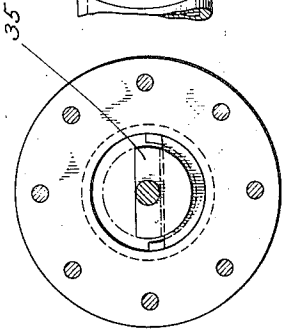


Fig. 10.

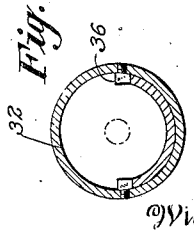


Fig. 9.

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Patented Aug. 3, 1926. 1,594,345

UNITED STATES PATENT OFFICE.

HERMAN E. BAKKEN AND WILLIAM G. HARVEY, OF NIAGARA FALLS, NEW YORK, ASSIGNORS TO AMERICAN MAGNESIUM CORPORATION, OF NIAGARA FALLS, NEW YORK, A CORPORATION OF NEW YORK.

PRODUCTION OF MAGNESIUM.

Application filed May 8, 1923. Serial No. 637,587.

Our invention relates to the production of metallic magnesium by distillation and has particularly in view the refining of crude magnesium which may be produced by the electrolytic decomposition of the fused salts of the metal. Magnesium, when produced from ores containing other metals and metalloids, is likely to be contaminated with them and the impurities should preferably be removed by some refining process prior to the use of the metal in the mechanical arts, for it is found that the presence of even small percentages of certain ingredients has an objectionable effect on the physical and chemical properties of the metal.

At atmospheric pressures magnesium boils at about 1120 degrees C., at which temperature distillation is impracticable owing to the fact that commercial apparatus will not withstand such high temperature operating conditions for any considerable period of time.

In carrying out our process, we utilize a high degree of vacuum which possesses peculiar advantages in connection with the production of pure metal. Magnesium is exceedingly active chemically, and slightly below its melting point, 651 degrees C. will combine with both oxygen and nitrogen of the air. Removal of air from the container in which the process is being carried out is therefore necessary in order to avoid the formation of magnesium compounds. Lowering of the boiling point by vacuum also has the advantage that it brings the operation within the temperature limits of commercial apparatus.

While the production of magnesium by distillation has been attempted before, the failure to control and maintain proper pressures and temperatures has not only caused rapid deterioration of the apparatus employed, but has also led to the refined metal being produced largely in the form of dust, in which form it is not available for most uses, as the dust can not readily be made to coalesce to produce solid metal. In the distillation process we employ a high degree of vacuum, approximately 4 mm. residual pressure, and a temperature of approximately 675 degrees C. By employing such a vacuum, two things are gained; it permits carrying out the distillation process at a much reduced temperature and it permits also a change in the physical state of a very oxidizable metal with a minimum formation of impurities.

Our process is capable of being carried out in a number of different types of apparatus; the one herein shown and described is to be considered as only illustrative of the invention, although it has been found to be practical in construction and operation and efficient in use.

Referring to the drawings for a more complete disclosure of the invention:

Fig. 1 is a vertical section through the furnace and tapping pot;

Fig. 2 is a plan view of the tapping pot with fragmentary section;

Fig. 3 is an elevation of the upper part of the pot;

Fig. 4 is a longitudinal section on the line 4—4 of Fig. 2;

Fig. 5 is a longitudinal section of the feeding device;

Fig. 6 is a sectional detail of the sealing mechanism for the top of the tapping pot;

Fig. 7 is a section on the line 7—7 of Fig. 2;

Fig. 8 is a section on the line 8—8 of Fig. 3;

Fig. 9 is a section on the line 9—9 of Fig. 5;

Fig. 10 is a section on the line 10—10 of Fig. 5;

Fig. 11 is a detail of the cover of the feeding mechanism, and

Fig. 12 is a section on the line 12—12 of Fig. 5.

In the drawings, 1 represents an iron retort made of extra heavy pipe, provided with closed ends and strengthened by the addition of reinforcing rings 2 welded integral therewith. A horizontal baffle plate 3 is provided in the upper portion of the retort and has openings 31 along the sides thereof to allow passage of the metallic vapors but to prevent the mechanical carrying over of solid or liquid particles with the said vapors. A feeding or charging device may be welded to one end of the retort and through this end solid metal may be charged into the retort at the beginning of the run; or if the run is to be continuous the metal can be placed inside of the retort piece by piece by using a feeding device as shown in

Fig. 5 and without breaking the vacuum on the main retort. If a batch run is to be made rather than a continuous run, the charging device is omitted and a blank flange is used instead for closing the retort.

Rising from the inner end of the retort is the delivery pipe 4 which connects the retort with the condenser 5, the said condenser consisting of a piece of extra heavy iron pipe standing vertically and having a sloping bottom 6 welded thereon. Joining the said condenser at its lowest point is a tapping pipe 7 which at its other end leads in a tapping pot 8.

The delivery pipe 4 opens into a condenser at about the middle point of the latter, and a deflector or baffle 32 may be arranged in the condenser just above the delivery pipe so as to throw the incoming vapors downward. The upper portion of the said condenser is closed by a removable cover 14 which can be maintained gas tight by any of the well known means and provides a construction by which the condenser may be opened and any solidified residue removed.

Connected to the condenser below the upper portion thereof is a vacuum pipe 9 leading to the vacuum pump by means of which the system is maintained under vacuum.

Three pyrometer wells are provided in the condenser at various heights, one pyrometer 10 being in the bottom to register the temperature of the molten metal contained therein; a second pyrometer 11 above the molten metal level, but below the delivery pipe opening so as to record the temperature of the incoming uncondensed vapor and a third pyrometer 12 above the delivery pipe and below the vacuum pipe so as to furnish an alarm signal, as it were, of any uncondensed vapor rising toward the outlet or vacuum pipe. In addition to these three pyrometers, there is another pyrometer 13 which is inserted in the lower part of the retort at the feeding end, thus registering the temperature of the boiling liquid magnesium metal. This pyrometer by a substantial rise in temperature gives a very definite indication when the metal has been removed from the retort by distillation.

The tapping pipe 7 is welded to the lower extremity of the condenser and projects horizontally through the brickwork of the furnace and is welded to the tapping pot by a return inside bend, as shown in Fig. 8.

The tapping chamber which is entirely outside of the brickwork of the furnace, rests upon the floor and is not heated. It is, however, connected into the vacuum system and during the operation of the furnace is under the same vacuum as the other parts of the apparatus. The tapping chamber need only be large enough to hold the cast iron mold into which the distilled metal is tapped. To permit removal of the mold, a removable cover 15 is provided to which is secured a rubber gasket 16 for making the joint gas tight. Hand screws 17 engage the top of the cover 15 and a cross bar 18, the latter carrying pivotal hooks 20, engaging the flange 19 on the body of the pot.

On the side of the tapping pot opposite to the side where the tapping pipe enters is a stuffing box 21 through which a tapping rod may pass to mechanically open the tapping pipe should it become clogged during a run.

The tapping pot has a compartment 22 on one of its sides, which is provided with a glass plate 23 tightly gasketed and provides a means for examining the interior of the pot at any time and is especially useful in noting conditions when a tap takes place. Electric lights 24 are provided on the inside of the pot for illuminating the interior.

Both the retort and the condenser are surrounded with brickwork with the exception of the upper part of the condenser and the outer extremity of the charging pipe. The brickwork which surrounds the retort is built so as to provide sufficient support for the retort and also sufficient space for the combustion of the fuel which is used for heating, and is supported on a steel frame 25, with a masonry arch 26 sprung over the retort, thus making it possible for portions of the brickwork to be removed for inspection and replacement of the retort, without necessitating the entire destruction of the furnace.

The products of combustion discharge through openings in the brickwork above the retort, a portion thereof being brought through an opening into the condenser compartment, thus passing around and keeping the delivery pipe at the proper temperature.

The brickwork surrounding the condenser is of similar construction to that surrounding the retort, providing the proper mechanical support for the condenser and also providing proper space for the combustion of the gases. This space is divided into two compartments which may be heated by two separate burners 27 and 28, so that either the lower portion of the condenser, which in practice would contain the molten metal, or the upper portion of the condenser, which contains the condensing vapors, can be independently heated.

The evacuation of the retort, condenser, tapping pot and necessary connecting pipes is accomplished by means of a vacuum pump capable of lowering the pressure at least to 1 mm.

For charging the retort, when a continuous run is being effected, a charging device, as shown in Fig. 5, is fastened to the charging end of the retort, in place of the blank flange which is placed on this end

in a batch run. This device comprises a cylindrical pipe 32, having a reciprocating gate valve 29, a trough 30 which is reciprocatable in the pipe and a removable cover 31 through which the metal may be placed in the trough. Slidably mounted in the front end of the charging device through a stuffing box 34 is a push rod 33 having a semicircular disc 35 on its end. For limiting the inward movement of the trough, a stop 36 carried by the inside of the pipe 32 engages a projection 37 on the upper end of the trough. In operation, when the vacuum on the charging device is equal to the vacuum on the retort, the valve 29 is opened, connecting the retort with the charging device. The rod 33 is now pushed forward to slide the trough forward until limited by the stop 36. The push rod is now rotated so that the disc 37 fits into the trough, which is pushed forward so that it pushes the metal pieces into the retort. After the charge has been fed in, the push rod 33 and trough 30 are withdrawn to their original positions, the valve 29 closed, the vacuum released on the charging device and the cover 31 removed, whereupon the parts are in position for repetition of the cycle.

Operation.

The process of volatilizing the metal may be carried out either intermittently or continuously. If intermittent, a definite amount of metal is added to the retort when in its open condition. It is closed tightly and the steps then follow of evacuation, heating, distilling over the batch, collecting it in the condenser, tapping, cooling down, opening the retort to remove the residue, and opening the tapping pot to remove the cast product.

A continuous run is one in which an original amount of metal is charged into the retort when in its open condition, the retort closed, evacuated, heated and while distillation is in progress, there is added at definite intervals, a stated amount of metal by means of the charging device shown in Fig. 5 so as to keep up the level of the molten liquid layer in the retort, and successive tappings of the condenser are made to remove the distilled product.

In starting the process after the furnace has been initially charged with metal, the system is evacuated until there is a residual pressure of approximately 2 to 10 mm. At this rarefaction, there is not a sufficient amount of oxygen or nitrogen present to unite with an appreciable amount of magnesium, and it is accordingly satisfactory for the process of distilling magnesium.

After the evacuation is complete, the burner under the retort having been started, the temperature is gradually raised until at about 140 degrees C., moisture and certain occluded gases in the solid metal are driven off. Heating is then continued until an average temperature of about 675 degrees C. is attained, at which temperature the magnesium is liquid and boils readily. In the apparatus described the condenser should be hot before any magnesium vapors are allowed to enter it from the retort so as to condense the vapors to a liquid rather than a solid. If the apparatus is designed to permit the removal of solid magnesium from the condenser, the condenser may be kept below the melting point of magnesium. If the condenser is too cold the condensed magnesium may deposit in a powdery form, which is ordinarily considered disadvantageous. The burners on the condenser are therefore lighted to keep it at the proper temperature.

The regulation of the temperature of the condenser is an important feature. If it is too hot at the exit, the vapors will not condense. At the bottom, in the apparatus illustrated the temperature should be regulated so as to keep the molten magnesium in a liquid condition so that tapping may be made possible and easy. There must always be a difference in temperature between the condenser and the retort so as to cause a definite flow of vapor from the retort to the condenser.

To tap the charge, an electric current of high amperage and low voltage is caused to flow through the pipe 7. The pipe, being at the lowest level of the condenser, has filled with magnesium metal after the first portions of the vapor condense, and the metal has solidified. The heat derived from the current melts the solid magnesium and permits the contents of the condenser to flow through the tapping pipe into the mold. As soon as the flow stops, the current is shut off to prevent the pipe 7 from being overheated.

The vapor pressure of magnesium at its melting point has not been determined by us experimentally, nor so far as we are aware has it ever been so determined. By various methods of interpolation and computation from known data, values from 1.15 to about 2 mm. have been derived for this factor. When the residual pressure in the retort is equal to the theoretical value of the vapor pressure of magnesium at its melting point, then the boiling point of magnesium is approximately 650 degrees C. which is substantially the melting point of the metal. At about 10 mm. pressure, the boiling point is about 750 degrees C. Within this range of pressures, the amount of residual air left in the retort is so small as to form no appreciable amount of magnesium compounds, and the process of distillation when carried out under such condition will produce magnesium of a high degree of purity,

and will permit an operating temperature that is commercially practicable.

We claim:

1. The method of producing substantially pure magnesium comprising subjecting material containing metallic magnesium to an absolute pressure slightly in excess of the vapor pressure of magnesium at its melting point and heating the said material to such a temperature as will cause the magnesium to boil, and thereafter reducing the temperature of said vapor to cause condensation of the magnesium.

2. The method of refining crude magnesium comprising distilling the liquid metal under an absolute pressure of approximately 2 to 10 mm. and thereafter causing said vapor to condense in a suitable form.

In testimony whereof we affix our signatures.

HERMAN E. BAKKEN.
WILLIAM G. HARVEY.